
Real-Time Health Monitoring
Personalized AI Analysis
Environment Monitoring
Multi-Factor Tracking
Early Risk Detection
NeuroTrace
Vibration Alerts
Care Taker Alert
Preventive Guidence
Emergency Detection
Privacy & Online Support

## HARDWARE LAYER
(Wearable Device)



**EEG SENSOR**
(Stress,neural activity)

**Temperature Sensor**
(Body & Environment)

**Heart Rate Sensor**
(HR & HRV)

**SpO2 Sensor**
(Oxygen level)

**Environmental Sensors**
(temperature,light, humidity,noise)

## EDGE LAYER
(On-Device Processing)

**Microcontroller**
(ESP32/similar)

- Collects sensor data
- Filters & Preprocesses
- Runs lightweight ML

(early detection)

**LOCAL STORAGE**
(On Device)

- Temporary data buffer
- Offline support

## SOFTWARE LAYER
(Mobile Application)

**USER INTERFACE**
- View headache history
- Track triggers & patterns
- Receive recommendations

**DATA VISUALISATION**
- Graphs & Insights
- Daily/weeks reports

**USER INPUT**
Text
- Headache Intensity
- Notes/mood/lifestyle

**SECURE AUTHENTICATION**
- Login/Signup
- Data Privacy & consent

(HTTPS)

## CLOUD & BACKEND LAYER
(Server Infrastructure)

**API Gateway**
- Handle requests
- Authentication
- Data routing

**BACKEND SERVICES**
- Business logic
- ML interactions
- User management

**DATABASE**
(Mongo DB)
- User data
- Sensor data
- Headache records
- Recommendations

## AI & ML LAYER
(Analysis & Predictions)

**DATA PROCESSING**
- Clean & Normalize
- Feature extraction
- Multimodel fusion

**ML MODEL**
- Pattern recognisation
- Trigger identification
- Risk Prediction

**Recommendation Engine**
- Personlized Suggestions
- Preventive actions
- Lifestyle guidence

## USER LAYER
(End User)



View Insights & alerts

Follow Recommendations

Manage lifestyle & triggers

Improve well being